Research Statement

Introduction

Figure 1: P-AgBot operating in cornfields

According to the "Critical And Emerging Technology List Update" by the U.S. National Science and Technology Council, agricultural and robotics technologies are recognized as critical and emerging areas, highlighting their importance for national technological leadership and security [1]. Robotics has enormous potential to impact precision agriculture positively. Motivated to realize this potential, I am dedicated to conducting research in the area of **agricultural robotics** to enhance precision agriculture with the goal of producing more food with finite resources while sustaining our planet. From my master's studies about the development of high-definition (HD) map update algorithms for autonomous vehicles at Korea University, I strengthened my expertise in localization, mapping, and perception in robotics, core foundations that support my research on agricultural robotics [2, 3]. I have developed novel technologies for **a robust and reliable mobile robotic system, Purdue-AgBot (P-AgBot), designed for autonomous navigation and agricultural task execution in large-scale cornfields** at the Internet of Things for Precision Agriculture (IoT4Ag), a U.S. NSF Engineering Research Center, with four partner universities located in diverse geographical regions: Purdue University, University of Pennsylvania, University of California, Merced, and University of Florida. My commitment to developing agricultural robotic systems has led me to focus on three components:

1. **Localization:** Reliable localization is a fundamental requirement for agricultural robotics, particularly for safe navigation and consistent, repeatable crop monitoring. Without robust localizers, autonomy breaks down, as it leads to navigation failure and inefficient task execution.
2. **Autonomous Navigation:** To perform agricultural tasks, robots must navigate to target positions intelligently and safely. In under-canopy environments, navigation methods using predefined paths are insufficient and unreliable due to Global Positioning System (GPS) limitations and environmental variability, e.g., plant growth and the change of soil conditions over time.
3. **Crop Monitoring:** Robotics and sensing technologies are essential in precision agriculture to automate tasks, such as crop sampling, monitoring, and remote sensing, that were traditionally performed manually, with enhanced efficiency. In agriculture, these tasks are essential to optimize resource use, increase productivity, and enhance sustainability.

In the following sections, I summarize my contributions in each component and present my future vision for advancing agricultural robotics in the Department of Biological and Agricultural Engineering at North Carolina State University.

Previous Works

1. Localization [4, 5]

Under canopies in cornfields, as shown in Fig. 1, GPS signals are unreliable due to cluttered vegetation, so that GPS-based localization systems, widely used in Unmanned Aerial Vehicles (UAVs) or tractors, cannot be utilized for under-canopy robots.

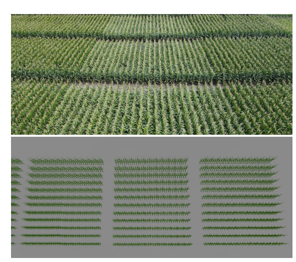

Figure 2: Testbeds: (Top) real cornfields at Purdue ACRE; (Bottom) simulated cornfield environment in ROS Gazebo

Generally, to solve the localization problems in GPS-denied environments, such as indoor or tunnel environments, vision or Light Detection and Ranging (LiDAR) sensors are widely used. However, in under-canopy cornfield environments, due to the varied illuminations, vision-based localization methods have shown degraded performance. State-of-the-art LiDAR-based odometry methods often fail to estimate robot poses in cornfields because these semi-structured or dynamic environments make it difficult to extract widely used simple geometric features, i.e., edge or planar features, from the surroundings, which are used to compensate for drifts in the existing approaches.

To address this challenge, I developed a LiDAR-based Simultaneous Localization And Mapping (SLAM) framework, P-AgSLAM [4], designed for in-row and under-canopy localization in cornfields through sensor fusion of wheel encoder from Unmanned Ground Vehicles (UGVs), an Inertial Measurement Unit (IMU), LiDAR, and GPS. A main contribution of this work is the design of a LiDAR odometry based on a novel feature extraction method that leverages morphological characteristics of cornfields, particularly corn stalks, rather than relying on the simple geometric features utilized in baseline methods. P-AgSLAM also takes advantage of relatively accurate GPS signals available when the robot is positioned outside crop rows or during early growth stages, by optionally fusing GPS measurements in a global pose optimization stage. To comprehensively assess the accuracy of pose estimation for each method, I conducted field tests in collaboration with the Purdue Agronomy and Agricultural & Biological Engineering groups at the Purdue Agronomy Center for Research and Education (ACRE) and built realistic simulated cornfield environments in the Robot Operating System (ROS) Gazebo, allowing system testing regardless of growing seasons, as illustrated in Fig. 2. The experimental results in [4] demonstrate that P-AgSLAM provides the most reliable framework for under-canopy mobile robot localization among the methods. On average, P-AgSLAM achieves nearly 90% lower final pose translation error compared to existing LiDAR-based methods. As a next step, to generalize the localization technology in agricultural environments, I proposed a novel agricultural-context-aware, feature-free LiDAR odometry framework that leverages a point-to-point Iterative Closest Point (ICP) formulation, a classical point cloud registration algorithm [5]. This framework improves the robustness of LiDAR odometry without relying on the performance of feature extraction in cluttered environments, outperforming all baseline methods with at least a 74% reduction in error. My research on localization for agricultural robots contributes to advancing the localization technology in semi-structured, dynamic, GPS-unreliable environments.

2. Autonomous Navigation [6]

Under-canopy cornfield environments present a unique challenge: cluttered hanging leaves from dense vegetation obstruct GPS signals and are often misidentified as non-passable obstacles by existing autonomous navigation and path planning systems. It prevents robots from driving through narrow rows and preserves crops.

Since vision-based approaches demonstrate degraded performance due to unstable perception under frequently changing illumination, and reliable GPS signals are inaccessible under canopies, 3D LiDAR becomes a more robust alternative for sensing the surroundings and enabling adaptive control for safe autonomous navigation. The key lies in effectively and efficiently processing LiDAR point clouds to ensure reliable navigation in cornfields, where dense vegetation generates highly noisy data.

Figure 3: Each stage of P-AgNav [6]

To achieve fully robotic automation in cornfields, I developed a fully autonomous navigation system, P-AgNav [6], based on 3D LiDAR range-view images. P-AgNav makes two main contributions: (1) by leveraging range-view images, it effectively captures meaningful features that represent the robot's facing contexts and enable adaptive controls; (2) it can reliably handle all navigation scenarios, including in-row navigation, out-row navigation, and row switching, using only a single 3D LiDAR without requiring GPS assistance or predefined waypoints. In addition, P-AgNav incorporates a collision avoidance module that protects critical crop structures for growth while safely navigating inside rows by continuously tracking row exits in the frames of range-view images. I validated P-AgNav through field trials in large-scale cornfields at Purdue ACRE and simulation. Across 25 trials with a total driving distance of 834.5 m, the (mean, standard deviation) of the number of collisions and human interrupts were (0.92, 1.41) and (1.88, 1.64), respectively. The experimental results demonstrate its robustness and highlight potential for scaling to a variety of row-crop environments.

3. Crop Monitoring [7–9]

Remote sensing technologies using UAVs and tractors are widely applied to monitor crops and improve productivity. However, these approaches show limitations in monitoring under canopies that contain critical indicators of crop health, such as leaf diseases, insect damage, and soil moisture stress. In addition, traditional manual monitoring is labor-intensive and time-consuming.

Given the size constraints of robot platforms to traverse narrow cornfield rows, it is not feasible to rely on a single configuration to monitor diverse metrics that have different characteristics. For example, a robotic arm with a cutter for physical leaf sampling is not efficient for measuring soil moisture. It motivated me to suggest modular robotic systems that perform various crop monitoring tasks autonomously. I focused on developing modules that track crucial agronomic metrics closely related to crop health and farm management.

It is critical to monitor soil moisture in agronomy, since accurate information on the moisture content of the soil reduces water resource wastage and prevents damage to crops due to over-irrigation. Therefore, I led a collaborative project with the University of Pennsylvania to develop an autonomous modular robotic system that drills into the soil, deploys biodegradable wireless inductor-capacitor sensors for passive soil moisture monitoring developed by their team, and collects real-time sensor data [9], as shown in Fig. 4a and 4b. Real-time monitoring of soil moisture through this robot eliminates traditional labor-intensive practices that rely on manual estimation and guessing. In addition to measuring soil moisture, as a means of yield prediction and stress detection in cornfields, I developed modules that accurately measure the height of individual corn and the radius of the stalk, metrics widely used in agronomy fields [7]. Vertically and horizontally mounted 3D LiDARs are used for height and stalk radius measurement, respectively, considering the field of view of each sensor for the corresponding metric. These modules enable the robot to automatically measure them with individual labels while traversing within rows. Beyond remote sensing or non-contact measurements, physical sampling is necessary in agronomy to obtain ground-truth and direct measurements of plant traits. I contributed to the development of deep learning-based leaf detection [8] and a physical sampling module [7] that employs a robotic arm with a customized nichrome wire cutter, as shown in Fig. 4c. This module has been extended to measure hyperspectral data and to incorporate optical leaf sensors through collaborations with other groups in IoT4Ag. This modular robotic system shows the present and future vision for scalable and automated crop monitoring for precision agriculture.

(a) (b) (c)

Figure 4: Modular robotic system for crop monitoring: (a) Drilling & Sensor deployment, (b) Soil moisture reader, (c) Physical sampling

These contributions have established a foundation for robust agricultural robotic systems in challenging environments. My research in agricultural robotics has also gained support for its potential adoption and commercialization of the technology and has been recognized through my selection as **a presenter in the Doctoral Consortium at IEEE International Conference on Robotics and Automation (ICRA) 2025**, as well as by receiving **Grant Shark Tank Award** and **the 2nd Place Poster Award at the U.S.-Korea Conference (UKC) 2025**. In addition, through the **U.S. NSF IoT4Ag AgBridge Summer Cohort Program**, I actively supported students in education and mentor teachers in understanding and applying our research on agricultural robotics by translating complex research concepts into accessible content.

Future Vision

From my research experience in developing agricultural robotic systems in large-scale cornfields, my future vision as a faculty member at North Carolina State University is to lead a research program that develops **generalizable and collaborative robotic systems for precision agriculture**. North Carolina State University has unique opportunities to

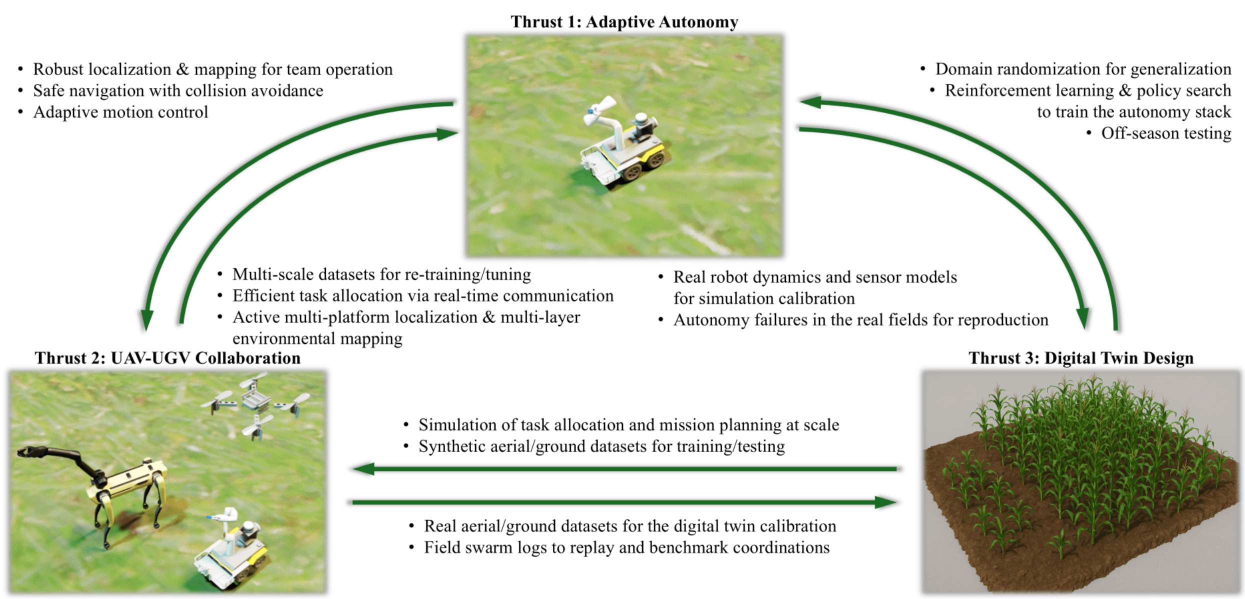

Figure 5: An illustration of my future research vision integrating three interconnected thrusts. The bidirectional links illustrate how each thrust will both influence and be influenced by the others to advance agricultural robotics.

advance this vision by integrating my expertise in robotics with North Carolina's strong ecosystem in agronomy, plant sciences, and climate-resilient agriculture. I am confident that this integration will achieve highly productive synergy across diverse disciplines, from robotics to agriculture. The research in my lab will ultimately sustain food production and address the global challenges related to labor shortages and finite resources in agriculture. In detail, my future research will pursue the following directions, which are interconnected with each other, with an illustrative overview in Fig. 5.

Thrust 1. Adaptive Autonomy for Diverse Agricultural Environments

To achieve precision agriculture from the perspective of robotics, scalability remains one of the most significant unsolved challenges, since agriculture, particularly in North Carolina spans a variety of domains, including corn, soybean, cotton, tobacco, and peanuts, and specialty crops such as sweet potatoes, apples, peaches, and other fruits, and each crop is characterized by distinct agronomical traits and growing conditions. My lab will develop an autonomy stack, which includes localization, navigation, perception, and manipulation, to enable agricultural robotic systems to adapt to temporal and domain variations and operate fully autonomously and robustly in diverse field configurations. The autonomy stack will leverage traditional multi-modal sensor fusion technologies (vision, LiDAR, IMU, and GPS) while integrating learning-based methods and foundation models to enhance adaptability and generalization across environments. Furthermore, through integration with the multi-platform systems that will be proposed in Thrust 2, this work will advance towards active multi-platform localization and multi-layer environmental mapping. For the design of safe navigation systems, the digital twin, which will be introduced in Thrust 3, will support reinforcement learning and adaptive policy training. I look forward to the deep collaborations for developing robust robot autonomy with Dr. Jaemin Lee in the Department of Mechanical and Aerospace Engineering. In parallel with adaptive autonomy algorithm design, I plan to collect and share open-source datasets based on field experiments in various agricultural research stations by collaborating with the College of Agriculture and Life Sciences at North Carolina State University, a land-grant institution, and through partnerships with farmers and collaborators across North Carolina. This plan is motivated by the fact that datasets consisting of calibrated and time-synchronized multi-modal sensor data collected over the growing season are highly valuable to the robotics research community, as agricultural field data are extremely limited. This work will contribute to broader accessibility and reproducibility in agricultural robotics research. This research aligns with the interests of multiple funding agencies, such as NSF CISE RI, NSF CPS programs in the U.S. For instance, the NSF CISE RI program supports research that aims to contribute new insights across the disciplinary areas in robust intelligence, which aligns with our research on developing adaptive and intelligent autonomy for diverse agricultural environments. In addition, I aim to establish international collaborations with the Rural Development Administration (RDA) and industrial partners such as Daedong Mobility Corporation in South Korea.

Thrust 2. UAV-UGV Collaboration for Large-Area, Multi-Scale Sensing

Achieving the full potential of precision agriculture requires multi-scale sensing with complete coverage of large-area environments, as it allows for a more comprehensive analysis of crop and field conditions and offers effective agronomic decision-making. My lab will design multi-platform systems, particularly focusing on collaborative UAV-UGV systems as a pathway toward large-area multi-scale sensing. In these systems, UAVs will cover a wide area from the sky, capturing hyperspectral imagery and canopy-level metrics, while UGVs will measure contact-level, under-canopy metrics such as disease detection and physical sampling. In Thrust 2, my lab will pursue two primary visions. (1) My lab will develop swarms of UAVs and UGVs capable of reliable communication between UAV-UGV, UAV-UAV, and UGV-UGV, enabling interactive exploration of optimal swarm coordination and active mission allocation for efficient field operations. To achieve this objective, robust and long-range communication systems are required, and I would be excited to form collaborations with Dr. Ismail Guvenc in the Department of Electrical and Computer Engineering, who has expertise in wireless communications and networking with UAVs. (2) My lab will explore methods to correlate aerial and ground measurements of the same metrics. For example, soil moisture measurements from two distinct approaches, UAV imagery and UGV wireless sensors, can be correlated, and the fusion of data from these different sources can provide more reliable and comprehensive analyses. I will leverage my expertise in modular robotic system design for the targeted crop monitoring. I would be excited to form collaborations with Dr. Ron Heiniger in the Department of Crop and Soil Sciences, who has expertise in remote sensing and precision agriculture, and this collaboration will synergize with my work to advance multi-robot systems for multi-scale sensing. This research will be strengthened by the autonomy technologies developed in Thrust 1 and validated through digital twin simulations in Thrust 3. This research aligns with the interests of multiple funding agencies, such as USDA NIFA SCRI, NSF CPS, and NRCS CIG. As an example, USDA NIFA SCRI focuses on new innovations and technology, including improved mechanization and technologies related to sustainable agriculture.

Thrust 3. Digital Twin Design for Off-Season Research and Generalization

Given the growing season constraints of each crop in North Carolina, digital twin environments will play a vital role in enabling off-season development and testing. Not only as an alternative to real testbeds during the off-season, digital twin technologies have also emerged as powerful tools for robot training and generalization across diverse fields. My lab will aim to design a digital twin calibrated with real-world data to realistically simulate farm environments and gain the ability to perform large-scale, unlimited robot operations. It will enable robust sim-to-real transfer for the research in Thrust 1 and 2 through seasonal field validation. These simulations will be built using widely adopted, off-the-shelf simulation platforms, such as NVIDIA Isaac Sim, Unreal Engine, and Unity. This direction will directly validate the proposed methods and drive the advancement of research in Thrust 1 and Thrust 2. In addition, there is currently a lack of open-access, high-fidelity agricultural simulation environments available for research and education. My lab's work will address this gap and contribute broadly to the agriculture and robotics societies by making the developed digital twin publicly available. Collaborations with Dr. Ron Heiniger and Dr. Jeff White in the Department of Crop and Soil Sciences, particularly on the modeling of crop and soil systems, can improve the environmental realism of digital twins. This research aligns with the interests of multiple funding agencies, such as NSF FRR programs, which support fundamental advances in robotics science, and ARPA-E OPEN solicitations.

These three thrusts in my future vision at North Carolina State University are deeply interconnected, each influencing and being influenced by the others, as illustrated in Fig. 5. This interconnectivity will establish a cohesive research agenda that advances agricultural robotics between autonomy, simulation, and multi-scale sensing. Under the strong partnership between North Carolina State University and the State of North Carolina, with the goal of addressing grand challenges of robotics, global food, and environment, I will foster active interdisciplinary collaborations and impactful research outcomes, as a faculty in the Department of Biological and Agricultural Engineering at North Carolina State University.

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